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CASE STUDY:1

TOPIC: HOSPITAL READMISSION RECORDS: DATA QUALITY AND WRANGLING

P24CS904
DATA SCIENCE

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HOSPITAL READMISSION RECORDS – DATA QUALITY AND WRANGLING

1. ABSTRACT

Healthcare organizations generate large amounts of patient information through hospital admissions, laboratory tests, medical procedures, diagnoses, and medication records. However, raw healthcare datasets often contain missing values, duplicate records, inconsistent categories, incorrect data formats, and extreme values. These data-quality problems can affect the reliability of healthcare analysis and decision-making.

This project focuses on data quality assessment and data wrangling using the Diabetes 130-US Hospitals dataset, which contains approximately 101,766 hospital encounter records collected from 130 hospitals in the United States between 1999 and 2008.

The main objective is to transform raw hospital readmission records into a clean, consistent, and structured dataset suitable for further healthcare analysis. The project uses Python libraries such as Pandas and NumPy to identify missing values, detect outliers, standardize categorical variables, normalize numerical attributes, and integrate diagnosis information using an ICD-9 lookup table.

The project also aims to restructure encounter-level records into a patient-level dataset, where each patient is represented by one row. Every preprocessing and cleaning decision will be documented to maintain transparency and reproducibility.

The final outcome will be a cleaned dataset, a Jupyter Notebook containing the complete preprocessing workflow, and a data-quality report describing the issues identified and the solutions applied.

Keywords: Healthcare Data Analytics, Data Quality, Data Wrangling, Missing Value Treatment, Outlier Detection, Data Normalization, ICD-9, Pandas, NumPy.

2. INTRODUCTION

Healthcare data plays an important role in improving patient care, hospital management, and medical decision-making. Hospitals maintain electronic health records containing patient demographics, admission details, medical diagnoses, laboratory results, medication information, and discharge records.

However, raw healthcare data is not always suitable for direct analysis. It may contain missing information, duplicate entries, inconsistent data representations, and abnormal numerical values.

For example:

- A patient's weight may not be recorded.
- Missing values may be represented using the symbol '?' instead of an empty cell.
- The same categorical information may have inconsistent representations.
- Numerical attributes may contain unusually high or low values.
- Diagnosis codes may be difficult to interpret without a reference table.

These problems can reduce the quality and reliability of data analysis.

Data cleaning and data wrangling are therefore essential steps in the healthcare data-processing pipeline.

This project uses the Diabetes 130-US Hospitals dataset to investigate data-quality issues and develop a systematic preprocessing workflow. The cleaned data will be organized at the patient level to make it easier to analyze patient histories and readmission-related information.

2.1 Domain

Healthcare Data Analytics.

2.2 Sustainable Development Goal

SDG 3 – Good Health and Well-being

This project supports SDG 3 by improving the quality and usability of healthcare data. Reliable healthcare information can support better analysis of patient histories and hospital readmission patterns.

3. PROBLEM STATEMENT

The Diabetes 130-US Hospitals dataset contains a large collection of hospital encounter records. Although the dataset provides valuable information about diabetic patients, it

contains several data-quality challenges that must be addressed before meaningful analysis can be performed.

The major challenges include:

1. Missing values represented using '?' codes.
2. Duplicate or repeated encounter records.
3. Outliers and abnormal numerical values.
4. Inconsistent categorical values.
5. Different numerical scales across attributes.
6. Diagnosis codes that require additional reference information.
7. Multiple hospital encounters associated with the same patient.

If these issues are not handled properly, they may lead to inaccurate summaries, misleading analysis, and unreliable conclusions.

Therefore, the problem addressed by this project is to assess, clean, standardize, integrate, and restructure the raw hospital readmission dataset into a reliable patient-level dataset while documenting every data-cleaning decision.

4. OBJECTIVES

The major objectives of this project are:

1. To examine the quality and structure of the Diabetes 130-US Hospitals dataset.
2. To identify missing values, duplicate records, and inconsistent data.
3. To handle missing values represented by '?' codes.
4. To identify and appropriately treat numerical outliers.
5. To standardize inconsistent categorical values.
6. To normalize selected numerical attributes into a common scale.
7. To integrate diagnosis information using an ICD-9 lookup table.
8. To transform encounter-level records into a patient-level dataset.
9. To document all preprocessing and cleaning decisions.
10. To generate a clean dataset suitable for further healthcare analytics.

5. DATASET DESCRIPTION

5.1 Dataset Details

The project uses the publicly available Diabetes 130-US Hospitals dataset from the UCI Machine Learning Repository.

Attribute	Description
Dataset name	Diabetes 130-US Hospitals
Data source	UCI Machine Learning Repository
Domain	Healthcare
Number of records	101,766
Number of hospitals	130
Time period	1999–2008
Features	47 listed features
Data format	CSV
Main purpose	Hospital readmission analysis

The dataset contains information about hospital encounters involving patients diagnosed with diabetes.

Each row represents a hospital encounter rather than a unique patient. Therefore, a patient may appear in multiple rows if they have been admitted to a hospital more than once.

5.2 Important Attributes

The dataset contains the following categories of information:

1. Patient Information

- patient_nbr
- race
- gender
- age
- weight

2. Hospital Admission Information

- encounter_id
- admission_type_id
- discharge_disposition_id
- admission_source_id
- time_in_hospital

3. Medical Information

- diag_1
- diag_2
- diag_3
- number of laboratory procedures
- number of medical procedures
- number of medications

4. Readmission Information

- readmitted

The readmission attribute contains three categories:

- NO – No readmission.
- greater than 30 – Readmission after 30 days.
- less than 30 – Readmission within 30 days.

5.3 Dataset Files

The project primarily uses two files:

1. **diabetic_data.csv:** Contains hospital encounter records.
2. **IDS_mapping.csv:** Provides descriptions for admission type, discharge disposition, and admission source identifiers.

An additional ICD-9 diagnosis reference table will be used to interpret the diagnosis codes stored in diag_1, diag_2, and diag_3.

6. PROPOSED METHODOLOGY

The proposed system follows a systematic data-cleaning and data-wrangling process.

6.1 Overall Workflow

Raw Dataset
↓
Data Loading
↓

Data Profiling and Quality Assessment
↓
Missing Value Identification and Treatment
↓
Duplicate Detection
↓
Outlier Detection and Treatment
↓
Categorical Data Standardization
↓
Numerical Data Normalization
↓
ICD-9 Lookup Integration
↓
Patient-Level Data Transformation
↓
Final Data Validation
↓
Cleaned Dataset and Data-Quality Report

6.2 Data Loading

The raw CSV dataset will be imported into the Python environment using Pandas.

The dataset will be examined to understand its structure, number of rows, columns, data types, and sample records.

Initial inspection will include:

- Dataset dimensions.
- Column names.
- Data types.
- Missing value counts.
- Duplicate record counts.
- Unique values in categorical attributes.

6.3 Data Quality Profiling

Data profiling is the process of examining a dataset to identify quality problems.

The following checks will be performed:

Quality Dimension	Verification
Completeness	Identify missing values
Uniqueness	Identify duplicate records
Validity	Check invalid values and ranges
Consistency	Check inconsistent categories
Accuracy indicators	Identify suspicious or implausible values
Data type integrity	Verify appropriate data types

The results will be documented before applying any cleaning operations.

6.4 Missing Value Treatment

Some missing values in the dataset are represented using '?'.

These symbols will first be converted into proper missing-value representations using Pandas.

The following steps will be performed:

1. Identify columns containing '?'.
2. Calculate missing-value counts and percentages.
3. Examine the importance of each affected column.
4. Select an appropriate missing-value treatment method.
5. Validate the cleaned values.

Possible treatment methods include:

- Removing records only when justified.
- Replacing numerical missing values using median or another suitable statistic.
- Replacing categorical missing values with 'Unknown'.
- Retaining missingness indicators where useful.

For example, if the weight attribute contains many missing values, its missingness will be reported instead of blindly filling every missing entry with an estimated value.

6.5 Duplicate Record Detection

Duplicate records can affect the accuracy of data analysis.

The project will identify:

- Completely duplicated rows.
- Repeated encounter identifiers.
- Multiple encounters belonging to the same patient.

Only confirmed duplicate encounter records will be removed.

Repeated patient identifiers will not automatically be treated as duplicates because a patient can have multiple genuine hospital admissions.

6.6 Outlier Detection and Treatment

Outliers are numerical values that differ significantly from the general distribution of a variable.

The project will use:

- Interquartile Range (IQR).
- Boxplot visualization.
- Minimum and maximum value checks.
- Domain-based validation.

The IQR method is calculated as follows:

IQR = Q3 - Q1

Lower Bound = Q1 - 1.5 × IQR

Upper Bound = Q3 + 1.5 × IQR

Values outside these boundaries will be flagged for investigation.

Outliers will not automatically be deleted. Valid but extreme healthcare observations will be retained wherever possible. Only invalid or clearly erroneous values will be corrected, excluded, or otherwise treated with a documented justification.

6.7 Categorical Data Standardization

Categorical attributes may contain inconsistent labels, unnecessary spaces, or invalid category representations.

The project will standardize:

- Race.
- Gender.
- Medical specialty.
- Admission categories.
- Discharge categories.
- Other relevant categorical fields.

For example, unnecessary leading and trailing spaces will be removed, and equivalent category labels will be consolidated.

Unknown and invalid categories will be handled separately from genuine categories wherever necessary.

6.8 Numerical Data Normalization

Numerical attributes may have different value ranges.

For example, the number of laboratory procedures and the number of medications may have different numerical distributions.

Min-Max normalization will be considered for selected numerical attributes.

The formula is:

Normalized Value = (X - Xmin) / (Xmax - Xmin)

This transforms selected values into a range between 0 and 1.

Normalization will be applied only to suitable numerical features. Patient identifiers, encounter identifiers, diagnosis codes, categorical IDs, and target labels will not be treated as continuous numerical measurements.

6.9 ICD-9 Lookup Integration

The diagnosis columns diag_1, diag_2, and diag_3 contain ICD-9 diagnosis codes.

An external ICD-9 reference table will be used to obtain the corresponding diagnosis descriptions.

The lookup integration process will involve:

1. Loading the ICD-9 reference table.
2. Examining diagnosis codes and descriptions.
3. Standardizing code formats.
4. Matching diagnosis codes with the reference table.
5. Adding diagnosis descriptions to the corresponding records.
6. Identifying unmatched diagnosis codes.

This integration will improve the interpretability of diagnosis information.

The IDS_mapping.csv file will separately be used to decode admission type, discharge disposition, and admission source identifiers.

6.10 Patient-Level Data Transformation

The original dataset contains multiple encounters for the same patient.

The project will transform the encounter-level dataset into a patient-level dataset containing one row per unique patient.

The patient_nbr attribute will be used as the grouping identifier.

Possible aggregation methods include:

Attribute	Aggregation Method
patient_nbr	Unique identifier
encounter_id	Encounter count or list
time_in_hospital	Mean, maximum, or total
num_lab_procedures	Mean or total
num_procedures	Mean or total
num_medications	Mean or maximum
diag_1, diag_2, diag_3	Diagnosis history
readmitted	Patient-level readmission summary

The aggregation strategy will be selected according to the meaning of each attribute.

For example, the number of hospital encounters can be calculated by counting the records associated with each patient.

Readmission status requires special consideration because a patient may have different readmission outcomes across multiple encounters. Therefore, the patient-level outcome will be explicitly defined and documented instead of selecting an arbitrary value.

6.11 Final Data Validation

After completing the preprocessing operations, the final dataset will be validated.

The validation process will verify:

- No unintended duplicate patient rows.
- Correct data types.
- Properly handled missing values.
- Consistent categorical values.
- Valid numerical ranges.
- Successful diagnosis lookup integration.
- One row per unique patient.
- Preservation of important clinical information.

7. TOOLS AND TECHNOLOGIES

The following tools and technologies will be used.

Tool	Purpose
Python	Main programming language
Pandas	Data loading, cleaning, grouping and merging
NumPy	Numerical calculations and missing-value handling
Jupyter Notebook	Interactive implementation and documentation
Matplotlib	Data visualization
Seaborn	Statistical visualization

7.1 Software Requirements

- Python 3.x
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

8. EXPECTED OUTCOMES

The proposed project is expected to produce the following outcomes:

1. Identification of missing values and data-quality issues.
2. A systematically cleaned hospital encounter dataset.
3. Proper treatment of missing and inconsistent values.
4. Detection and documented handling of numerical outliers.
5. Standardized categorical attributes.
6. Normalized selected numerical features.
7. Integration of ICD-9 diagnosis descriptions.
8. A patient-level dataset containing one row per unique patient.
9. A reproducible Jupyter Notebook containing all cleaning operations.
10. A four-page data-quality report describing the original problems, cleaning decisions, and final validation results.

The project will provide a structured and reusable preprocessing workflow for healthcare data analysis.

9. DELIVERABLES

The project will produce three major deliverables.

9.1 Cleaned Dataset

A processed CSV file containing cleaned and standardized patient-level healthcare records.

9.2 Jupyter Notebook

A documented notebook containing:

- Dataset loading.
- Data profiling.
- Missing-value analysis.
- Duplicate detection.
- Outlier analysis.
- Categorical standardization.
- Numerical normalization.

- ICD-9 lookup integration.
- Patient-level aggregation.
- Final validation.

9.3 Four-Page Data-Quality Report

The report will include:

- Original dataset characteristics.
- Identified data-quality problems.
- Missing-value and outlier analysis.
- Cleaning methods and justifications.
- Before-and-after quality measurements.
- Final validation summary.

10. CONCLUSION

This project proposes a systematic approach to improving the quality of healthcare data using the Diabetes 130-US Hospitals dataset.

The project focuses on identifying and handling missing values, duplicate records, outliers, inconsistent categories, and numerical scaling issues. It also integrates ICD-9 diagnosis information and transforms multiple hospital encounters into a patient-level representation.

By using Python, Pandas, NumPy, and Jupyter Notebook, the project aims to produce a reliable, consistent, and reusable healthcare dataset.

The final cleaned dataset and documented preprocessing workflow can serve as a foundation for future healthcare analytics, patient-history analysis, and hospital readmission studies.

11. REFERENCES

1. Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). *Diabetes 130-US Hospitals for Years 1999–2008*. UCI Machine Learning Repository. https://doi.org/10.24432/C5230J
2. Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., & Clore, J. N. (2014). Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records. *BioMed Research International, 2014*, Article 781670. https://doi.org/10.1155/2014/781670
3. McKinney, W. (2010). Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science Conference*, 56–61.
4. Harris, C. R., et al. (2020). Array programming with NumPy. *Nature, 585*, 357–362. https://doi.org/10.1038/s41586-020-2649-2
5. UCI Machine Learning Repository. *Diabetes 130-US Hospitals Dataset Documentation*. https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008